Towards better understanding of circulating fluidized bed boilers : a data mining approach

Paper Summary

SUMMARY PASSAGE 1

Introduction

From combustion point of view the main challenges for the existing boilers are caused by a wider fuel selection (increasing share of low quality fuels), increasing share of bio fuels, and co-combustion. In steady operation, combustion is affected by the disturbances in the feed-rate of the fuel and by the incomplete mixing of the fuel in the bed, which may cause changes in the burning rate, oxygen level and increase CO emissions. This is especially important, when considering the new biomass based fuels, which have increasingly been used to replace coal.

SUMMARY PASSAGE 2

Introduction

The development of a set of combined software tools intended for carrying out signal processing tasks with various types of signals has been undertaken at the University of JyvÃ¤skylÃ¤ in cooperation with VTT Processes 1 . This paper presents the vision of further collaboration aimed to facilitate intelligent analysis of time series data from circulating fluidized bed (CFB) sensors measurements, which would lead to better understanding of underlying processes in the CFB reactor.

SUMMARY PASSAGE 3

Addressing Business Needs With The Data Mining Approach

Currently there are three main topics in CFB combustion technology development; once through steam cycle, scale up (600 -800 MWe), and oxyfuel combustion.

SUMMARY PASSAGE 4

Addressing Business Needs With The Data Mining Approach

When the CFB boilers are becoming larger, not only the mechanical designs but also the understanding of the process and the process conditions affecting heat transfer, flow dynamics, carbon burnout, hydraulic flows etc. have been important factors. Regarding the furnace performance, the larger size increases the horizontal dimensions in the CFB furnace causing concerns on ineffective mixing of combustion air, fuel, and sorbent. Consequently, new approaches and tools are needed in developing and optimizing the CFB technology considering emissions, combustion process, and furnace scale-up [3].

SUMMARY PASSAGE 5

Fig. 1. A Simplified View Of A Cfb Boiler Operation With The Data Mining Approach

The produced heat converts water into steam that can be utilized for different purposes. The measurements from sensors S F , S A , S L , S H , S S and S E that correspond to different input and output parameters are collected in database repository together with other metadata describing process conditions for both offline and online analysis. Conducting experiments with pilot CFB reactor and collecting their results into database creates the necessary prerequisites for utilization of the vast amount of DM techniques aimed to identifying valid, novel, potentially useful, and ultimately understandable patterns in data that can be further utilized to facilitate process monitoring, process understanding, and process control.

SUMMARY PASSAGE 6

Fig. 1. A Simplified View Of A Cfb Boiler Operation With The Data Mining Approach

Currently, we are concentrated on estimation of the responses of the burning rate and fuel inventory to changes in fuel feeding. Different changes in the fuel feed, such as an impulse, step change, linear increase, and cyclic variation have been experimented with pilot CFB reactor. In [1] we focused on one particular task of estimating similarities in data streams from the pilot CFB reactor, estimating appropriateness of the most popular time-warping techniques to the particular domain.

SUMMARY PASSAGE 7

Concluding Remarks And Future Work

Those include development of techniques and software tools, which would help to monitor, control, and better understand the individual processes (utilizing domain knowledge about physical dependence between processes and meta-information about changing conditions when searching for patterns, and extracting features at individual and structural levels). Further step is learning to predict char load from few supervised examples that likely lead to adoption of active learning paradigm to time series context. Progressing in these directions we, finally, will be able to address the strategic needs related to construction of intelligent CFB boiler, which would be able to optimize the overall efficiency of combustion processes.

SUMMARY PASSAGE 8

Concluding Remarks And Future Work

Thus, a DM approach being applied for time series data being accumulated from CFB boilers can make critical advances addressing a varied set of the stated problems.